HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **analog and digital hardware** for servo-drive products, ensuring adherence to performance specifications and reliability targets.
- Supported the **hardware life cycle** from requirements definition through integration and testing as part of a multi-site development team.
- Participated in **design reviews and business reviews**, contributing technical insights and feedback to project stakeholders.
- Developed **Python scripts** to automate data validation and reporting tasks, processing approximately **1,500 records** per batch.
- Contributed to the development of **Java Spring Boot REST APIs** for backend services, focusing on feature enhancements and bug fixes.
- Managed **change-control processes** for system configuration updates, coordinating User Acceptance Testing and post-go-live support across interconnected systems.
- Containerized **Java applications** using **Docker** and deployed them to **AWS EC2** instances, managing build pipelines with **Jenkins**.

SKILLS

Hardware Design – Analog Electronics, Digital Electronics, Power Electronics, DFM, DFT

Embedded Systems – VHDL, Real-time Embedded Systems, FPGA Communication, A/D Converters, D/A Converters, Optical Encoders

Motor Control – Permanent Magnet Synchronous Motors, Power Conversion Principles, Thermal Management

Software Engineering – Java, Spring Boot, Python, SQL, REST APIs, Docker, Git, Jenkins, Linux

Cloud & Operations – AWS (EC2, S3, Lambda, CloudWatch), Production Support, SLA Management, Incident Response

Analytical & Soft Skills – Mathematical Skills, Analytical Skills, Problem Solving, Communication Skills, Interpersonal Skills, Teamwork, Independent Work

PROJECTS

Motor Controller Hardware Design (HCL Technologies)

- Designed and documented **digital control logic** for a motor controller product using **VHDL**, targeting a **FPGA**.
- Developed **Python scripts** to simulate and test **analog and power electronics** interfaces, verifying data integrity across approximately **500 test cases**.
- Collaborated with a multi-site team to integrate hardware designs into a product development lifecycle, ensuring adherence to **AS9100D** standards.

Embedded Systems Integration & Testing (HCL Technologies)

- Implemented and tested **hard real-time embedded system** components for a servo-drive product, achieving target deadlines for critical milestones.
- Developed **Python automation scripts** for testing device communication protocols with **A/D and D/A converters**, reducing manual testing time by **40%**.
- Supported integration efforts by debugging and resolving issues between hardware components and embedded firmware, contributing to a smoother UAT phase.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)